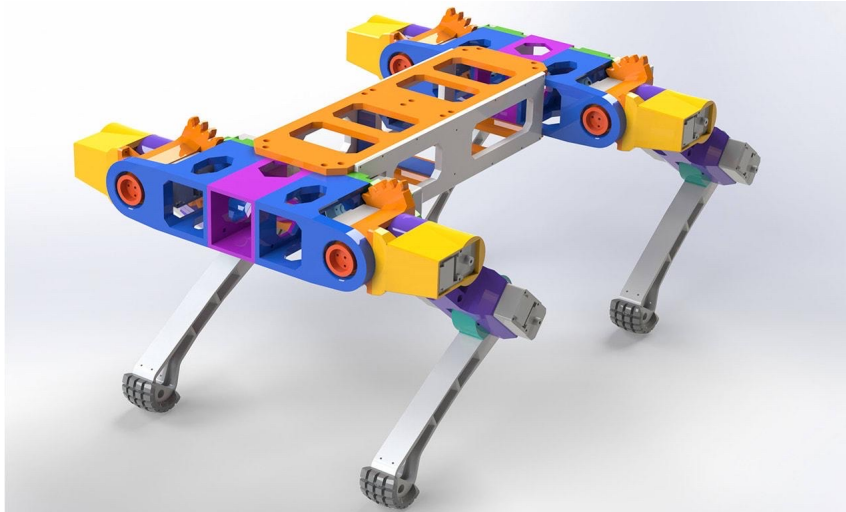


Design of a Robotic Dog

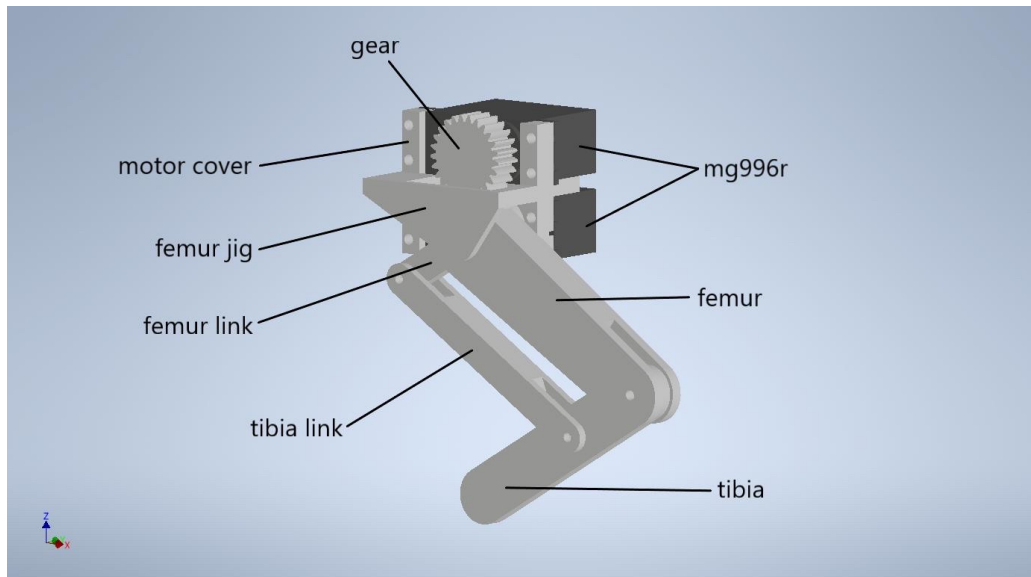


Body Shape and Frame



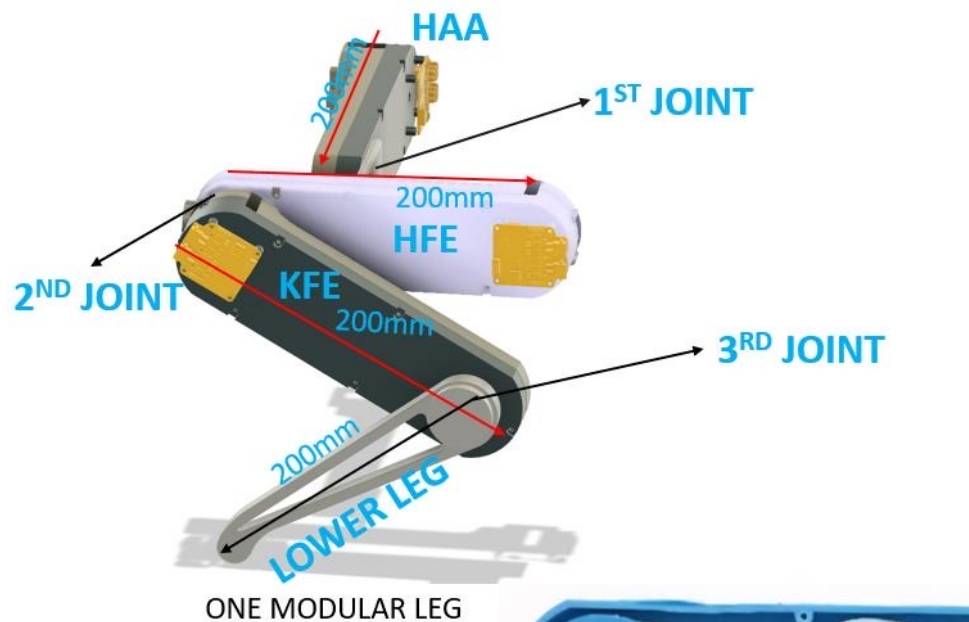
- The robot has a rectangular body.
- The frame is lightweight and rigid.
- The battery and controller are placed inside the body.
- The frame supports all robot components.

Leg Design



- Four identical legs.
- Each leg consists of three links.
- The legs provide balance during walking.

Joints & Degrees of Freedom



- Hip Joint
- Knee Joint
- Ankle Joint
- Each joint allows rotational movement.

Servo Motor



- Servo motors were selected.
- They provide accurate motion.
- Easy to control.
- Suitable for robot joints.

Torque Calculation

Weight = 1 kg

Force =

$$F = m \times g$$

$$F = 1 \times 9.81$$

$$F = 9.81 \text{ N}$$

Distance = 0.3 m

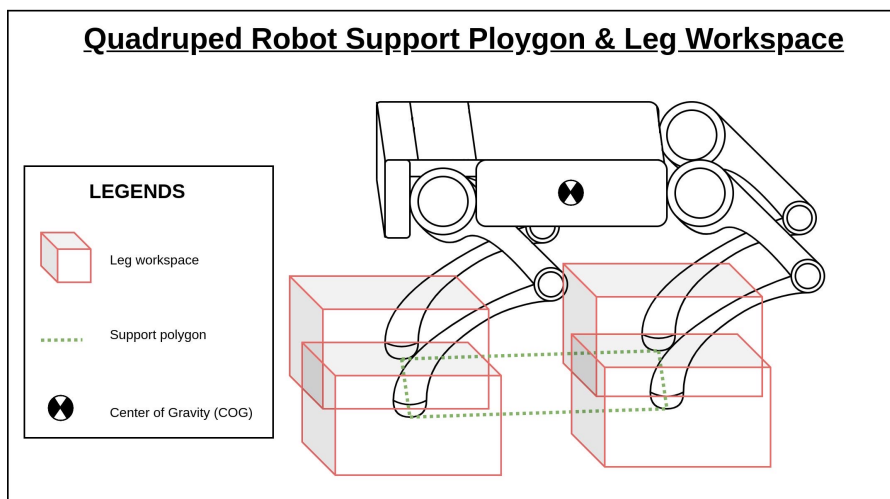
Torque

$$\tau = F \times r$$

$$\tau = 9.81 \times 0.3$$

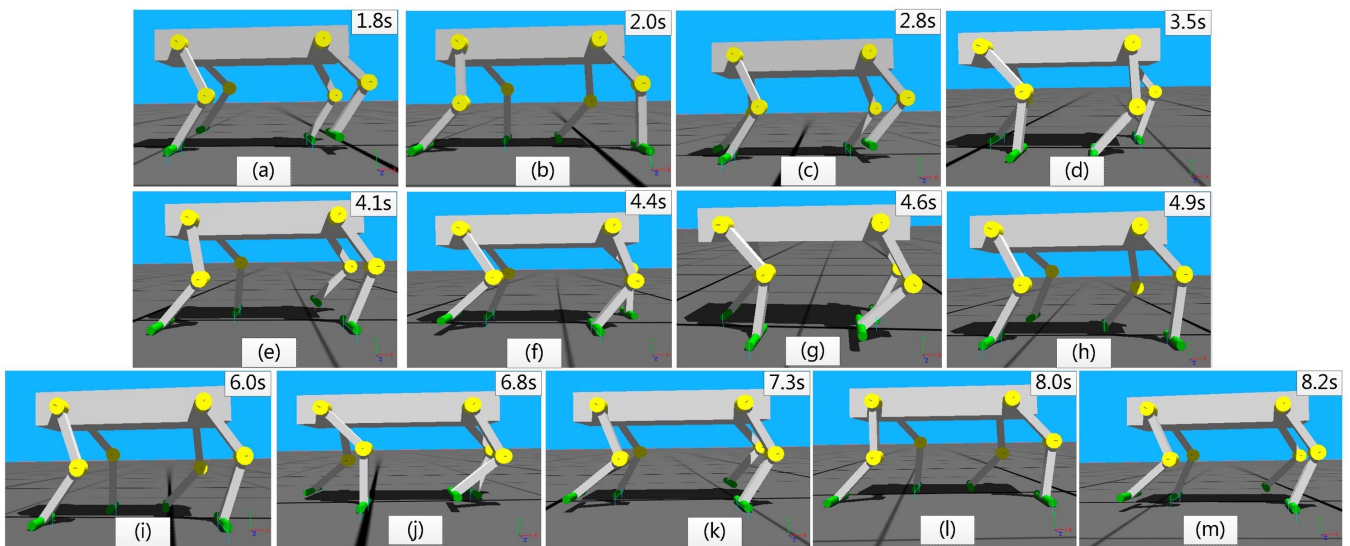
$$\tau = 2.94 \text{ N}\cdot\text{m}$$

Center of Gravity



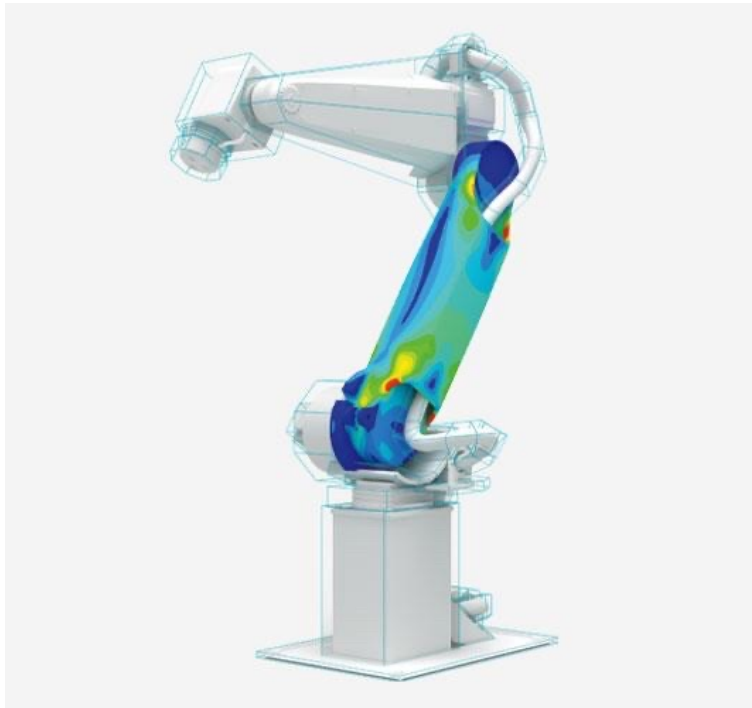
- The center of gravity should stay in the middle.
- Proper weight distribution improves stability.
- A low center of gravity prevents tipping.

Walking Method



- Crawl gait.
- One leg moves while the other three support the robot.
- This walking method provides high stability.

Mechanical Challenges



- Weak frame.
- Incorrect motor selection.
- Poor balance.
- High friction.
- Low torque.